

HAO DAI



Department of Computer Science
University College London
169 Euston Rd., London NW1 2AE

Mobile: +44-07709968630
Email: daihaovigg@gmail.com
 Google Scholar Github
 Homepage

RESEARCH INTERESTS

My research interests lie in **Generative Models**, **Efficient AI**, and **Continual Learning**, with a unified focus on developing scalable and adaptable multimodal generative systems across vision, audio, and video domains. My goal is to advance the foundation of general-purpose, efficient, and lifelong generative AI capable of consistent understanding and synthesis across diverse data modalities.

EDUCATION

Ph.D in Computer Science Sept. 2019 - May 2024
University of Chinese Academy of Sciences, China
• Thesis: Theories and Methods of Edge-Cloud Collaboration for Edge Intelligence
• Advisor: Prof. Yang Wang

Master of Electronic Engineering Sept. 2015 - May 2017
Wuhan University of Technology, China
• Thesis: Real-time Congestion Analysis System for Urban Rail Transit Based on Big Data
• Advisor: Prof. Feng Lv and Prof. Fan Zhang

B.S. in Electronic Engineering Sept. 2011 - May 2015
Wuhan University of Technology, China

RESEARCH EXPERIENCE

University College London **Research Fellow** May 2025 - Present

Research Projects:

- *CoVLA - Cloud-Edge Collaborative Data Generation Framework for VLA Learning*, NVIDIA Academic Grant Program, 2025.
- *MultiTasking and Continual Learning for Audio Sensing Tasks on Resource-Constrained Platforms*, EPSRC (EP/X01200X/1), 2023-2025.

University of Southampton **Research Fellow** Sept. 2024 - May 2025

Research Projects:

- *MultiTasking and Continual Learning for Audio Sensing Tasks on Resource-Constrained Platforms*, EPSRC (EP/X01200X/1), 2023-2025.

Shenzhen Institutes of Advanced Tech. **Research Assistant** Sept. 2019 - May 2024
Chinese Academy of Sciences

Research Projects:

- *Theory and Method of Hardware and Software Cooperative Optimization for Federated Learning*, Chinese General Program, 2023-2026.
- *Edge Cloud Collaborative Computing Methods and Applications in C-V2X*, Shenzhen-Hong Kong-Macau S&T Program (Category C), SGDX20220530111001003, 2023-2025.
- *Key Technology of Network Architecture Optimization in AI Computing Cluster*, Key-Area R&D Program of Guangdong Province (No. 2021B0101400005), 2021-2022.
- *Cloud Computing Architecture and Platform for Human-Computer Integration-Data Driven Technology for Self-Adaptive and Evolutionary Applications*, National Key R&D Program of China (No. 2018YFB1004804), 2018-2021.

SELECTED PUBLICATIONS

1. **Hao Dai**, and Jagmohan Chauhan, “**Continual Generalized Category Discovery: Learning and Forgetting from a Bayesian Perspective**”, Forty-Second International Conference on Machine Learning (ICML), 2025.
2. **Hao Dai**, Chong Tang, Jagmohan Chauhan, “**ViRN: Variational Inference and Distribution Trilateration for Long-Tailed Class-Incremental Learning**”, ICML Workshop on Machine Learning for Audio, 2025.
3. **Hao Dai**, Yuqing Zhu, Xueying Zhu, “**CoTraX: An Efficient Parallel Training Method for On-Policy Deep Reinforcement Learning**”, the Pacific Rim International Conference on Artificial Intelligence (PRICAI), 2025.
4. **Hao Dai**, Yang Wang, Jerome Yen, Yong Zhang, Chengzhong Xu, “**Cost-Efficient Sharing Algorithms for DNN Model Serving in Mobile Edge Networks**”, IEEE Transactions on Services Computing (IEEE TSC), 2023, vol. 16, no. 4, pp. 2517-2531.
5. **Hao Dai**, Jiashu Wu, André Brinkmann, Yang Wang, “**Neighborhood-oriented Decentralized Learning Communication in Multi-Agent System**”, 32nd International Conference on Artificial Neural Networks (ICANN), 2023.
6. **Hao Dai**, Jiashu Wu, Yang Wang, Chengzhong Xu, “**Towards Scalable and Efficient Deep-RL in Edge Computing : A Game-based Partition-based Approach**”, Journal of Parallel and Distributed Computing (JPDC), 2022, vol. 168, pp. 108-119.
7. **Hao Dai**, Yang Wang, Kenneth B. Kent, Lingfang Zeng, Chengzhong Xu, “**On Metadata Managements in Large-Scale Distributed File Systems—Scalability, Performance and Availability**”, IEEE Transactions on Parallel and Distributed Systems (IEEE TPDS), 2022, vol. 33, no. 12, pp. 3850-3869.
8. Yang Wang, **Hao Dai**, Xinxin Han, Pengfei Wang, Yong Zhang, Chengzhong Xu, “**Cost-Driven Data Caching in Content Delivery Edges**”, IEEE Transactions on Mobile Computing (IEEE TMC), 2023, vol. 22, no. 3, pp. 1384-1400.
9. Jiashu Wu, **Hao Dai**, Kenneth B. Kent, Jerome Yen, Chengzhong Xu, Yang Wang, “**Open Set Dandelion Network for IoT Intrusion Detection**”, ACM Transactions on Internet Technology (ACM TOIT), 2024. To appear.
10. Jiashu Wu, **Hao Dai**, Yang Wang, Yong Zhang, Dong Huang, Chengzhong Xu, “**PackCache: A Cost-driven Packable Model Caching Algorithm for Machine Learning in Distributed Clouds**”, IEEE Transactions on Computers (IEEE TC), 2023, vol. 72, no. 4, pp. 1208-1214.
11. Jiashu Wu, **Hao Dai**, Yang Wang, Kejiang Ye, Chengzhong Xu, “**Heterogeneous Domain Adaptation for IoT Intrusion Detection: A Geometric Graph Alignment Approach**”, IEEE Internet of Things Journal (IOTJ), 2023.

PAPERS UNDER REVIEW

1. **Hao Dai** and Jagmohan Chauhan, “TFMAudio: High-Fidelity Long-Form Text-to-Audio via Mamba-based Flow Matching”, under review.
2. Chong Tang, **Hao Dai**, Jagmohan Chauhan, “Parallax: Runtime Parallelization for Operator Fallbacks in Heterogeneous Edge Systems”, under review.

PROFESSIONAL ACTIVITY

Reviewer

- The International Conference on Learning Representations (ICLR 2026)
- The Annual Conference on Neural Information Processing Systems (NeurIPS 2025, 2023)
- IEEE Transactions on Parallel and Distributed Systems (2022)
- IEEE Transactions on Services Computing (2021)
- Journal of Parallel and Distributed Computing (2021)

Program Committee Member

- The 31th International European Conference on Parallel and Distributed Computing (2025)
- The 30th International European Conference on Parallel and Distributed Computing (2024)

TEACHING EXPERIENCE

Teaching Assistant

- | | | |
|-------------------------|--|-------------|
| • Design Sensor Systems | Dept. of Computing Science
University College London | Fall 2025 |
| • Distributed Storage | Dept. of Computing Science
Shenzhen University of Advanced Technology | Spring 2022 |
| • Operating System | Dept. of Computing Science
Shenzhen University of Advanced Technology | Fall 2021 |

AWARD AND HONORS

- | | |
|--|-----------|
| President Scholarship of Shenzhen Institute of Advanced Technology | 2022-2023 |
| Outstanding Student of University of Chinese Academy of Sciences | 2022-2023 |
| Outstanding Student of University of Chinese Academy of Sciences | 2021-2022 |
| University of Chinese Academy of Sciences Ph.D Scholarship | 2019-2022 |

WORK EXPERIENCE

Shenzhen Institute of Beidou Applied Technology

Data Mining Engineer

May 2016 - Aug. 2019

- Responsible for real-time processing and storage of petabyte-scale traffic data, and for developing deep learning-based data mining algorithms, event modeling, and knowledge graph extraction.

COMPUTER SKILLS

- | | | |
|---------------------------------|---------|--------------|
| Operating Systems: | Linux | 2012-Present |
| Programming Languages: | Java | 2011-Present |
| | C/C++ | 2011-Present |
| | Python | 2013-Present |
| | Scala | 2014-Present |
| Distributed Programming: | Spark | 2016-Present |
| | Pytorch | 2017-Present |
| | Ray | 2021-Present |
| | CUDA | 2022-Present |
| | Jax | 2022-Present |
| Development Tools: | Git | 2016-Present |
| | Docker | 2019-Present |